EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 2, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-03 **Date:** August 13, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted for final grading)
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

---

1. How many different hexadecimal strings of length 7 are there? Each position in a hexadecimal string can contain any of the digits (0–9) or the letter symbols (A–F). Show the necessary steps for computation. **[CO2, C2] [Mark: 5]**

2. The PhD program in a university has 27 professors. How many graduate students must be enrolled simultaneously to ensure that there exists one professor who is advising at least 12 graduate students? You should assume that no graduate student is allowed to enroll without an advisor.

[CO2, C2, Mark: 5]

EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 2, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-09 **Date:** August 14, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted for final grading)
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

---

1. A user on a computer system has a password. The password must be 7 characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? **[CO2, C2] [Mark: 5]**

2. A box contains 20 marbles in 2 different colors (10 + 10). What is the smallest number of marbles that must be taken out to guarantee that at least 4 marbles of the same color are chosen? Justify your answer. **[CO2, C2, Mark: 5]**

EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 2, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-10 **Date:** August 14, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 10 (5 will be counted for final grading)
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

---

1. A license plate consists of 3 uppercase English letters followed by 4 decimal digits. How many different license plates can be formed if repetition of letters and digits is allowed? Show the necessary steps for computation. **[CO2, C2] [Mark: 5]**

2. A drawer contains 30 balls in 4 different colors $(10+10+5+5)$. What is the smallest number of balls that must be taken out to guarantee that at least 5 balls of the same color are chosen? Justify your answer. **[CO2, C2, Mark: 5]**